INAMIGOS FOUNDATION • EDUCATIONAL VISION DOCUMENT

WHY INAMIGOS FOUNDATION ASSIGNED THE AI WEBSITE GENERATION TASK

UNDERSTANDING THE EDUCATIONAL VISION BEHIND THE INTERNSHIP

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PLATFORM CASE STUDY

SECONDLIFE

AI Circular Economy Platform

EXECUTIVE SUMMARY

As artificial intelligence reshapes the tech landscape, the role of a software engineer is shifting from manual coding to system design and collaboration with AI. Recognizing this change, the **InAmigos Foundation** structured the website generation assignment to give students direct experience with this new way of working.

Practical Application Over Theory

Instead of just studying theory, interns are tasked with using AI to plan, design, and deploy a web application. This practical approach requires students to think about system requirements, user needs, and deployment logistics, preparing them for the demands of the modern tech industry.

AI-Assisted Web Development

Learning to collaborate with AI tools is becoming a key skill for developers. By using AI to design layouts, generate assets, and structure code, interns learn how to build and validate ideas quickly, improving both their development speed and resourcefulness.

This document details the educational objectives of the assignment and shows how **SecondLife**, an AI-powered circular economy platform, was developed to meet these learning goals while building a useful, community-focused tool.

TASK OBJECTIVES

The website generation task is structured around key learning goals designed to help interns grow technically and creatively:

AI-Assisted Engineering

Learn to use AI code helpers to design interfaces, write backend logic, and deploy code quickly.

Improve Creative Thinking

Solve complex problems by designing unique layouts and interactions that make web pages feel engaging.

UI/UX Design Practices

Learn about layout typography, color theory, and responsive grid design to build easy-to-use interfaces.

Website Architecture

Practice structuring assets, setting up folders, and managing component dependencies systematically.

Modern Git Workflows

Build confidence deploying applications to production hosts like Vercel and managing code using GitHub.

Portfolio Projects

Develop high-quality, fully functional projects to showcase technical skills to future employers.

WHY WEB GENERATION MATTERS

A website serves as an organization's digital identity. Every business, NGO, and creative needs a clear web presence to reach their audience and share their message:

Rapid Idea Prototyping

Using AI to generate web layouts speeds up prototype development. This allows teams to validate ideas, gather user feedback, and test models in hours rather than weeks, making it easier to refine product direction early.

Combining Tech & Communication

Building websites teaches developers how to balance technical code with visual layouts and user copy. This helps engineers learn how to communicate technical concepts clearly to non-technical users.

Improving Global Accessibility

Understanding modern web standards ensures pages load quickly, run smoothly on mobile devices, and remain accessible to all users, regardless of their location or hardware.

By teaching these web development fundamentals, the assignment prepares students to build clean, functional, and accessible digital products.

WHY AI WAS INCLUDED

Rather than bypassing manual development, using artificial intelligence in this assignment helps developers build more complex, interactive platforms more efficiently:

Productivity & Fast Prototyping

AI assistants handle repetitive tasks like writing CSS utility classes or setting up boilerplate files. This saves developers time, letting them focus on key features like custom audio synthesis or complex state routers.

Encouraging Innovation

With AI assistance, developers can experiment with advanced web features—like WebGL particles or custom audio APIs—that they might have avoided due to steep learning curves, encouraging more creative solutions.

Collaboration as a Key Skill

Knowing how to work with AI models (e.g., using effective prompts and debugging code output) is becoming a core skill. This assignment helps students learn how to use AI to solve technical challenges.

AI changes the developer's role from writing simple code to managing complex systems, prompting students to think like product engineers.

HOW SECONDLIFE ALIGNS

Instead of building a simple portfolio, I developed **SecondLife**—an AI-driven circular economy platform designed to address solid waste and support community donation logistics.

AI-Driven Circularity

The platform uses computer vision to sort items into four sustainability paths: **Repair** (via local technicians), **Donation** (to verified NGOs), **Recycling** (to local centers), or **Resale** (to reuse markets).

Practical Social Good

SecondLife connects users directly with local charities and repair shops, using digital dashboards to show how individual recycling choices reduce landfill waste and carbon emissions.

Immersive User Experience

By combining Three.js 3D globes, custom cursor movements, and Web Audio API sound cues, the site makes ecological action feel rewarding and engaging.

SECONDLIFE APPLICATION DASHBOARD

[3D WebGL Globe] • [AI Scanning Portal] • [ESG Calculator]

SKILLS DEMONSTRATED

Developing SecondLife required applying a variety of technical and creative skills:

Frontend & 3D Web

Built responsive, state-managed layouts using Next.js, and integrated Three.js particle systems for visual globes.

`Next.js 16` `Three.js` `R3F`

Audio Synthesis

Programmed synthesized sound effects using W3C Web Audio APIs, creating click and hover cues client-side.

`Web Audio API` `LFO Modulation`

AI Collaboration

Used AI coding assistants to speed up interface design, write styling classes, and debug code errors.

`Prompt Engineering` `IDE Copilot`

Git & Deployment

Managed code updates using Git, and set up continuous integration pipelines to deploy the site to Vercel.

`GitHub` `Vercel CI/CD`

Design Thinking & UX

Created unified glassmorphic card grids, responsive columns, and micro-interactions to deliver a polished, accessible interface.

LEARNING OUTCOMES

The internship provided valuable experience in managing software development, user interface design, and product documentation:

1. Practical AI Collaboration

Learned how to collaborate with AI coding helpers, evaluate output code quality, and integrate AI assistance with manual coding tasks to build features more efficiently.

2. Design Thinking and Prototyping

Developed wireframes, designed responsive layouts, and structured components to turn a high-level idea into a functional platform for sustainable resource management.

3. Professional Communication and Documentation

Practiced explaining technical designs and architecture flows through clear project reports, preparing me to work with design teams and product managers.

These lessons helped bridge the gap between classroom theory and the skills needed to design, develop, and document real-world software products.

FUTURE VISION & REFLECTION

SecondLife provides a starting point for a scalable circular economy platform. Future versions could integrate several advanced technologies:

Computer Vision & Local Networks

Integrating active machine learning models will allow the platform to identify materials and wears automatically, while expanding the partner network will coordinate pickups and recycling logistics across regions.

IoT Bins & Eco-Rewards

Integrating smart disposal bins will verify item deposits automatically, allowing users to track their environmental impact and earn eco-points for circular choices.

Ultimately, this task shows that web development is about more than writing code—it is an opportunity to design solutions for real-world environmental and social challenges.

"Technology creates its greatest value when it improves lives and protects our future."

THANK YOU

I appreciate the opportunity to work on this assignment. Building this project helped me develop practical web development skills and design a functional platform for sustainable circular habits.

GitHub Codebase

melvindeepesh-boop/SECOND-LIFE

Live Application

second-kohl-ten.vercel.app

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